IBScovars: reference-paper-style application

Simultaneous statistical inferences of treatment–control mean differences

Methods / dose comparison	Point estimate (treatment − control)	95% simultaneous confidence interval
Proposed method with shrinkage		
1 vs. 0	0.260	(-0.102, 0.622)
2 vs. 0	0.327	(-0.040, 0.694)
3 vs. 0	0.184	(-0.179, 0.547)
4 vs. 0	0.302	(-0.078, 0.681)
Proposed method without shrinkage		
1 vs. 0	0.260	(-0.100, 0.621)
2 vs. 0	0.327	(-0.030, 0.684)
3 vs. 0	0.184	(-0.174, 0.543)
4 vs. 0	0.302	(-0.085, 0.688)
Dunnett's procedure with naive case deletion		
1 vs. 0	0.292	(-0.067, 0.651)
2 vs. 0	0.326	(-0.010, 0.662)
3 vs. 0	0.208	(-0.137, 0.554)
4 vs. 0	0.305	(-0.070, 0.679)

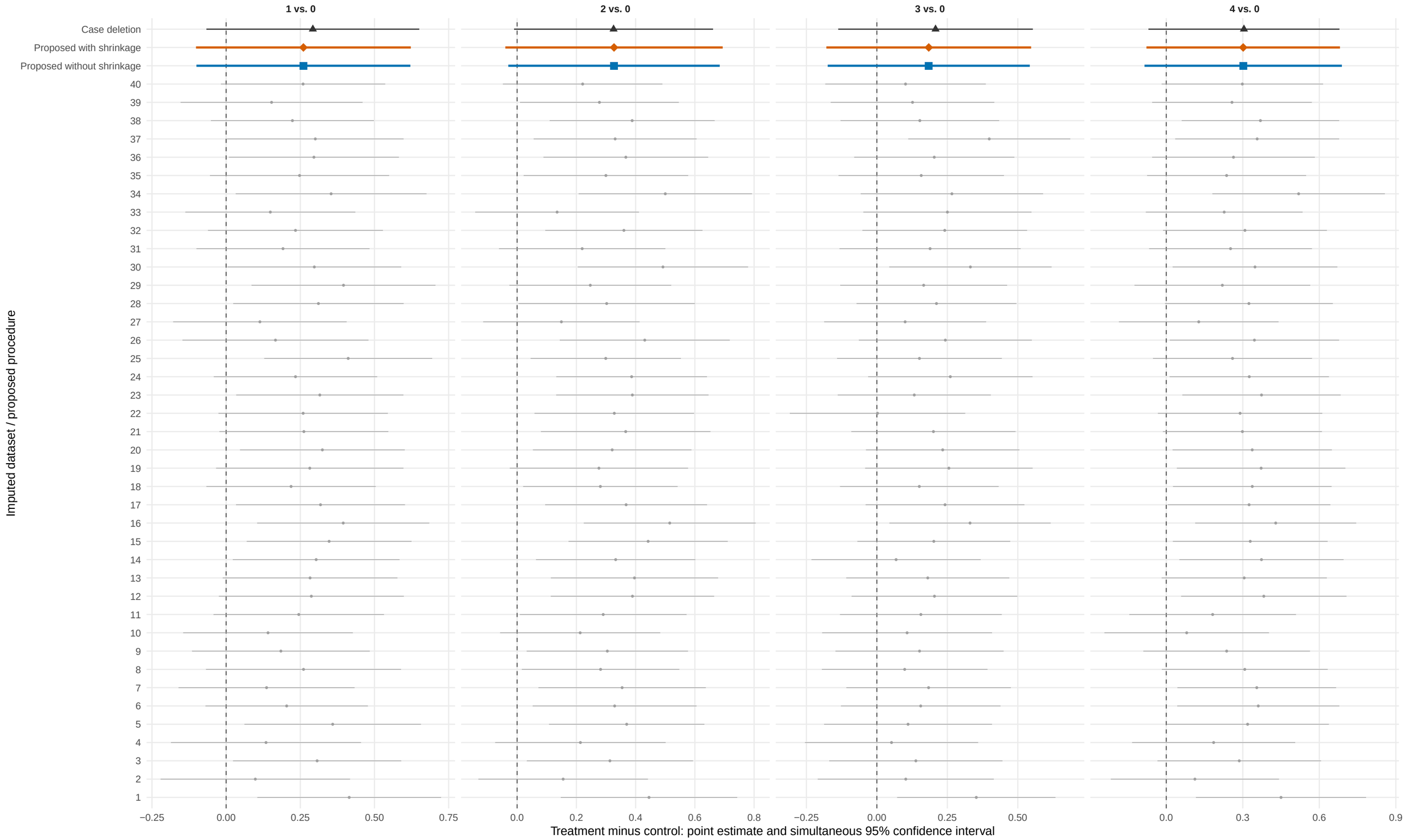
One application analysis; 40 regression imputations as specified by the uploaded script. The target missingness is 30% MAR; the realized missingness is 118/369 = 31.98%.

Shrinkage and non-shrinkage share the same 40 imputations and point estimates. Naive case deletion is the comparator explicitly included in the supplied script. No additional simulation loop or imputation-method comparison was added.

The accompanying forest plot reproduces the script’s reference-paper layout: 40 individual completed-dataset results, both proposed procedures after Rubin combination of the 40 imputations, and naive case deletion, separately for each dose contrast.

IBScovars: regression imputation − individual imputations and proposed procedures

40 regression imputations; 30% target MAR (31.98% realized missingness)



Highlighted rows show the final proposed procedures using all 40 regression imputations. Grey numbered rows are individual imputations.
Rubin combination is used for both procedures; this is not an equal−variance assumption. Highlighting does not indicate superiority.